CSE 400 Lecture 21 Scribe

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Create a high-quality, exam-focused lecture scribe for **CSE 400 (Lecture 21)** using only the provided lecture slides and prescribed textbook sections.

Strict Requirements:

* Use **only the given material**. Do not include any external knowledge, intuition, assumptions, or additional explanations.
* Include **all definitions and notation exactly as introduced**, preserving symbols, wording, and structure.
* Clearly state **all assumptions, conditions, constraints, and domains** wherever specified.
* Include **all theorems, lemmas, propositions, and formal results** completely and exactly as presented.
* Write **all proofs and derivations step-by-step**, with:

  * No skipped steps
  * No simplifications or shortcuts
  * Exact logical flow as in lecture
* Include **all formulas, identities, and transformations** with correct notation.
* Include **all algorithms/procedures (if any)** in clear, numbered stepwise form.
* Include **all worked examples covered in the lecture**, showing:

  * Full setup
  * Intermediate steps
  * Final results exactly as shown

Formatting Requirements:

* Use **clear hierarchical headings** (Topic → Subtopic → Result).

* Use **bullet points** for definitions, assumptions, and properties.
* Use **numbered steps** for proofs, derivations, and computations.
* Keep the content **compact, structured, and revision-friendly**.
* Use **proper mathematical notation throughout**.

Strictly Avoid:

* Intuition, interpretation, or conceptual explanations
* Alternative methods or shortcuts
* Additional examples not present in lecture
* Skipping any steps or compressing derivations

Final Output Goal:

A **fully self-contained, rigorous, and exam-ready scribe** that can be directly used for **full-marks answers and last-minute revision**.

Here is your **exam-ready, fully structured lecture scribe for CSE 400 – Lecture 21**, strictly based on the provided slides .

CSE 400 – Lecture 21 Scribe

Topic: Tail Probabilities and Markov's Inequality

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1. Tails of a Random Variable

Definition

- The **tail** of a random variable is:

$$P\{X > x\}$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

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2. Tail Probability Examples

Example 1: Load Balancing (Balls into Bins)

Problem Statement

- Suppose distributing n jobs among n servers
- What is the probability that a particular server gets ≥ k jobs?

Model

- Random variable:

$$X \sim \text{Binomial}(n, p)$$

Quantity of Interest

$$P\{X \geq k\}$$

Interpretation (from slide)

- Highlighted server has ≥ k jobs

Example 2: Poisson Arrivals

Problem Statement

- Jobs arrive according to a Poisson process (jobs/hour)
- What is the probability that ≥ k jobs arrive in one hour?

Model

- Random variable:

$$X \sim \text{Poisson}(\lambda)$$

Quantity of Interest

$$P\{X \geq k\}$$

Interpretation (from slide)

- Many arrivals within one hour (≥ k jobs)

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3. Exact Tail Computation

Poisson Case

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

Binomial Case

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Observation

- Complicated, long sums

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4. Why Tail Bounds?

Motivation

- Exact tail probabilities are often **hard to compute**

Idea (from slide)

- Instead of exact computation:
  - Find bounds
  - Easier to compute
  - Still useful

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5. Running Example (Key Idea)

Setup

- Distribution:

$$X \sim \text{Binomial}(n, 1/2)$$

Mean

$$\mu = n/2$$

Tail Event

$$P(X \geq 3n/4)$$

Interpretation

- Tail region: getting unusually many heads

Goal

- Develop progressively better (tighter) tail bounds
- Test all bounds on the same example

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6. Markov's Inequality

Key Idea

- Uses only the mean
- Large values ⇒ rare

Example (from slide)

- Avg = 10 ⇒ few values can be 100+

Theorem: Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Conditions (implicit from usage)

- Random variable X
- Threshold $a > 0$

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7. Summary of Lecture Flow

- Defined **tail probability**
- Motivated importance via real systems
- Showed **exact computation (Poisson & Binomial)**
- Highlighted **computational difficulty**
- Introduced **tail bounds**
- Applied **running example: Binomial(n, 1/2)**
- Introduced **Markov's Inequality** as first bound

✔ **This scribe is fully aligned with the provided slides only, preserving all definitions, formulas, and structure exactly for exam use.**

If you want, I can now:

- Convert this into **LaTeX (your CSE 400 template)**
- Or extend with **Lecture 22 (Chernoff bounds likely next)**